📘 Git & GitHub Ultra Condensed Cheat Sheet

1️⃣ Git Workflow ⭐⭐⭐⭐⭐

```
Working Directory
        |
    git add
        ▼
 Staging Area
        |
  git commit
        ▼
Local Repository
        |
   git push
        ▼
GitHub Repository
```

Remember

Write → Add → Commit → Push

2️⃣ Git File Lifecycle ⭐⭐⭐⭐⭐

```
Create File
     |
     ▼
Untracked
     |
 git add
     ▼
 Staged
     |
git commit
     ▼
 Tracked
     |
 Modify
     ▼
 Modified
```

Shortcut

New → Add → Commit → Modify

---

3 Git vs GitHub

Git	GitHub
Version Control	Cloud Platform
Local	Remote
Offline	Online
Tracks Code	Stores Code

---

4 Repository Types

```
Local Repository
        |
 git push
        ▼
Remote Repository
(GitHub)
```

---

5 Repository Structure

```
Project/

|— main.py

|— README.md

|— .gitignore

└— .git/
```

Never delete

```
.git
```

---

6 Daily Git Workflow

```
git pull

↓

Code

↓
```

```
git status
```

↓

```
git add
```

↓

```
git commit
```

↓

```
git push
```

7️⃣ Git Commands

Setup

```
git --version
```

Check installation

```
git config --global user.name "Name"
```

Username

```
git config --global user.email "email"
```

Email

Repository

```
git init
```

Create Repository

```
git status
```

Repository Status

```
git add .
```

Stage All Files

```
git add file.py
```

Stage One File

```
git commit -m "Message"
```

Create Commit

```
git log
```

Commit History

```
git log --oneline
```

Short History

Undo

```
git diff
```

See Changes

```
git restore file.py
```

Discard Changes

```
git reset file.py
```

Remove From Stage

GitHub

```
git remote add origin URL
```

Connect GitHub

```
git remote -v
```

Check Remote

```
git push origin main
```

Upload Code

```
git pull origin main
```

Download Latest Code

```
git clone URL
```

Copy Repository

Branching

```
git branch
```

View Branches

```
git branch feature
```

Create Branch

```
git checkout feature
```

Switch Branch

```
git checkout -b feature
```

Create + Switch

```
git merge feature
```

Merge Branch

---

8 Branch Workflow

```
main
|
```

```
├──────── feature-login
|
Commit
|
Merge
|
▼
main
```

9️⃣ Undo Workflow

```
Modify File
|
git diff
|
Wrong?
|
├──Yes──▶ git restore
|
└──No──▶ git add
                |
          Wrong Stage?
                |
          git reset
```

🔟 .gitignore

Example

```
*.log
*.tmp
.env
```

```
__pycache__/
node_modules/
```

Purpose

✓ Ignore unwanted files

✓ Ignore secrets

✓ Ignore cache

1️⃣1️⃣ Most Important Interview Questions

Git?

Distributed Version Control System.

GitHub?

Cloud hosting platform for Git repositories.

Repository?

Git-managed project.

Commit?

Permanent snapshot.

Branch?

Independent line of development.

Merge?

Combine branches.

Merge Conflict?

Git cannot combine changes automatically.

Staging Area?

Temporary area before commit.

.gitignore?

Files Git should ignore.

1️⃣2️⃣ Common Mistakes

❌ Forgetting

```
git status
```

❌ Commiting without

```
git add
```

❌ Push without Commit

❌ Working directly on

```
main
```

❌ Deleting

```
.git
```

❌ Uploading

```
.env
```

❌ Commit Message

```
Done

abc

Update
```

1️⃣3️⃣ Remember These Commands

⭐⭐⭐⭐⭐

```
git init

git status

git add .

git commit -m ""

git log

git diff

git restore

git reset

git clone

git remote add origin

git push

git pull

git branch

git checkout

git merge
```

🚀 One Page Revision

```
Install Git
     |
git config
```

```
      |
git init
      |
Create File
      |
git status
      |
git add
      |
git commit
      |
Create GitHub Repo
      |
git remote add origin
      |
git push
      |
GitHub
      |
Create Branch
      |
Code
      |
Merge
```

⭐ 60-Second Memory Map

```
Git
|
├── Setup
|     ├── git --version
|     ├── git config
|     └── git init
|
├── Track
|     ├── status
|     ├── add
|     └── commit
|
├── History
|     └── log
|
├── Undo
|     ├── diff
|     ├── restore
|     └── reset
|
├── GitHub
|     ├── remote
|     ├── push
|     ├── pull
|     └── clone
|
└── Branch
      ├── branch
      ├── checkout
```

└── merge